Lihsuan Chuang

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Chapel Hill, NC-27510, United States

Research Interests

My research interest spans across computer architecture and security with a focus on microarchitectural side-channel attacks. My current work explores keystroke inference attacks on Arm-based hardware and high-performance microarchitectural weird machines.

Education

- **University of North Carolina at Chapel Hill** — *09/2024 – 06/2026*
  Master of Science in Computer Science — Advisor: Prof. Andrew Kwong — Chapel Hill, United States
- **University of British Columbia** — *09/2020 – 06/2024*
  Bachelor of Applied Science in Computer Engineering with Distinction — Vancouver, Canada

Publications

- **KeyTAR: Practical Keystroke Timing Attacks and Input Reconstruction** — *09/2024 – 09/2025*
  Mufan Qiu*, **Lihsuan Chuang***, Dohhyun Kim, Huaizhi Qu, Tianlong Chen, Andrew Kwong — *Accepted at IEEE S&P 2026*
  - Designed and demonstrated the **first end-to-end input-reconstruction attack** from inter-keystroke timing signals, effective in both native and browser environments
  - Built a dataset collection pipeline to replay typing samples and capture high-fidelity side-channel traces at scale
  - Implemented the first general Prime+Probe keystroke extraction attack without the need of a target

Awards & Fellowships

- **IEEE S&P Student Travel Grant** — *04/2026*
  IEEE Symposium on Security and Privacy 2026 — San Francisco, United States
- **UBC Four-Year Doctoral Fellowship** — *03/2026*
  Awarded upon PhD Admission — University of British Columbia — Vancouver, Canada
- **UBC Outstanding International Student Award** — *03/2020*
  Awarded upon Bachelor Admission — University of British Columbia — Vancouver, Canada

Invited Talks & Presentations

- **IEEE Symposium on Security and Privacy 2026 — KeyTAR** — *05/2026*
  Conference Proceedings Presentation — San Francisco, United States
- **UNC COMP 590-184 Hardware Security — KeyTAR** — *03/2026*
  Guest Lecture — Chapel Hill, United States
- **UNC COMP 560 Artificial Intelligence — KeyTAR** — *02/2026*
  Guest Lecture — Chapel Hill, United States

Research & Professional Experience

- **University of North Carolina at Chapel Hill — Dept. of Computer Science** — *09/2024 – present*
  Graduate Research Assistant, Advisor: Prof. Andrew Kwong — Chapel Hill, United States
  - Led the full research cycle for *KeyTAR*: threat modeling, dataset collection, attack implementation, and evaluation against real-world targets
  - Investigated into the feasibility of keystroke extraction attacks on Arm-based hardware devices
- **IEEE Symposium on Security and Privacy 2026 [web]** — *08/2025, 05/2026*
  External Reviewer and Presenter — Oakland, United States
  - Reviewed submissions in microarchitectural security and side-channel attacks
  - Provided detailed feedback on organization, scientific value, novelty, and technical correctness
  - Presented *KeyTAR* to an international audience of security researchers at one of the premier venues in computer security (IEEE S&P)
- **ACM Conference on Computer and Communications Security 2025 [web]** — *10/2025*
  Attendee — Taipei, Taiwan
  - Participated in workshops and presentations of published works
  - Exchanged research ideas with security researchers from globally renowned institutions

- **UBC Thunderbots** [web] *09/2021 – 06/2024*
  Software Engineer Vancouver, Canada
  - Supported integration of a physics-accurate game simulator
  - Developed test suites validating the effectiveness of robot tactics
  - Implemented a fault-detection and robot-substitution algorithm
- **iNet Internship** *06/2021 – 08/2021*
  Web Developer Taipei, Taiwan
  - Designed interactive web pages integrating popular third-party widgets
  - Performed systematic testing to surface edge-case bugs from unexpected user behavior

Teaching Experience

- **University of North Carolina at Chapel Hill – COMP 590-184: Hardware Security** *01/2026 – 05/2026*
  Teaching Assistant, Advisor: Prof. Andrew Kwong Chapel Hill, United States
  - Responded to student technical inquiries on Campuswire
  - Hosted weekly in-person office hours covering hardware security concepts
  - Delivered two guest lectures on keystroke timing side-channel attacks, bridging course theory with cutting-edge published research
- **University of British Columbia – Department of Electrical and Computer Engineering** *09/2022 – 06/2024*
  Teaching Assistant (APSC 160, CPEN 211, CPEN 212) Vancouver, Canada
  - Led tutorial sessions reinforcing core concepts in introductory programming and digital systems design
  - Graded coursework and exams to support instructor course management
  - Collaborated with instructors to improve overall course delivery and assessment quality

Selected Projects

- **KeyTAR: Practical Keystroke Timing Attacks and Input Reconstruction** *09/2024–06/2025*
  Tools: C, Javascript, Python, NodeJS [code]
  - Implemented an automated cache trace collection tool on native and the browser environment
- **HistoData: Automated Trading System and Visualization Tool** *09/2023–06/2024*
  Tools: C#, ReactJS, Python, NodeJS, SQL, Interactive Broker API [code]
  - Implemented an automated trading data visualization desktop tool
  - Built middleware to bridge a ReactJS front end to the Interactive Broker API
- **SDMS: Smart Doorbell System** *01/2022–05/2022*
  Tools: Verilog, Lua, Flutter, AWS [code]
  - Integrated hardware, mobile application, and cloud server for a smart doorbell
  - Established cross-team communication protocols to ensure cohesive system integration
- **RISC Computer** *01/2021–05/2021*
  Tools: DE1-SOC & Verilog [code]
  - Designed and implemented a RISC processor (ALU, registers, datapath, memory) capable of executing a subset of ARMv7 instructions using HDL

Technical Skills

- **Programming Languages:** C, C++, Java, Python, R, Matlab, Verilog, JavaScript, HTML, CSS
- **Security & Systems:** Side-channel analysis, microarchitectural attacks, hardware security, embedded systems, HDL design
- **Languages:** Mandarin Chinese (Native), English (Native), Japanese (JLPT N2, 12/2024), French (Intermediate)